ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-3 **WINTER SEMESTER, 2023-2024**

DURATION: 30 MINUTES **FULL MARKS: 15**

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. The zero y-intercept version of the linear regression model for predicting the response for a given set of n data points (x_1, y_1), (x_2, y_2), $\dots$, (x_n, y_n) is $3+2$ (CO4) (PO1)

$$y = a_1 x$$

where a_1 is the only parameter of the regression model.

i. Derive the formula for finding the optimal value of a_1.

ii. Prove that the value of a_1 obtained using the formula derived in Question-1(i) is unique and corresponds to the absolute minimum of the optimization criterion used.

Solution:

i. The residual error for each given data point is,

$$E_i = y_i - a_1 x_i$$

So the sum of the square of the residual errors is,

$$S_r = \sum_{i=1}^{n} E_i^2 = \sum_{i=1}^{n} (y_i - a_1 x_i)^2$$

To find a_1, we look for the value of a_1 for which S_r is the absolute minimum. At first, let's do the first derivative test.

$$\frac{dS_r}{da_1} = 2\sum_{i=1}^{n}(y_i - a_1 x_i)(-x_i) = 0$$

$$\Rightarrow -2\sum_{i=1}^{n} y_i x_i + 2\sum_{i=1}^{n} a_1 x_i^2 = 0 \Rightarrow -2\sum_{i=1}^{n} y_i x_i + 2a_1\sum_{i=1}^{n} x_i^2 = 0$$

$$\Rightarrow a_1 = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} x_i^2}$$

Rubric:

- 0.5 point for the correct equation of E_i.
- 1 point for the correct equation of S_r.
- 1.5 points for the correct derivation of a_1.

ii. To prove the uniqueness of a_1, we need to prove that its value corresponds to the absolute minimum of S_r. For this, let's invoke the second derivative test.

$$\begin{aligned}\frac{d^2S_r}{da_1^2} &= \frac{d}{da_1}\left(\frac{dS_r}{da_1}\right)\\ &= \frac{d}{da_1}\left(2\sum_{i=1}^{n}(y_i - a_1x_i)(-x_i)\right)\\ &= \frac{d}{da_1}\left(\sum_{i=1}^{n}(-2x_iy_i + 2a_1x_i^2)\right)\\ &= \sum_{i=1}^{n}2x_i^2 > 0\end{aligned}$$

by pragmatically assuming that all the x_i values are not zero at the same time. Hence, S_r yields the absolute minimum value at that value of a_1. So the value of a_1 that is obtained using the formula in Question-1(i) is unique. ☐ Q.E.D.

Rubric:

- 1.5 points for the correct second derivative test.
- 0.5 point for the correct conclusion.

2. Suppose you are given a set of n data points (x_1, y_1), (x_2, y_2), $\ldots$, (x_n, y_n) and you want to regress the data to an mth order Polynomial model, where $m < n$. How would you choose a suitable value of m? 3 (CO3) (PO1)

Solution:

In order to choose a suitable value of m, we need to maintain a Bias-Variance tradeoff. If the value of m is too high, the polynomial model will overfit the given n data points and result in high variance. On the other hand, if the value of m is too small, the polynomial model will underfit the given n data points and result in high bias.

3. Suppose, you are monitoring the progress of a homogeneous chemical reaction in order to calculate the rate constant k and the order of the reaction n. The rate law expression for the reaction can be represented with the Power model 5 (CO2) (PO1)

$$-r = kC^n$$

where r is the reaction rate and C is the concentration of the reactant. The values of reaction rate $-r$ and the respective concentration C of the reactant that you have calculated over the course of time are shown in the table below —

Determine the value of the rate constant k and the order of the reaction n using the data provided in Table-1. You are allowed to apply data transformation.

Solution:

Concentration, C ($gmolL^{-1}$)	**Reaction Rate,** $-r$ ($gmolL^{-1}s^{-1}$)
4	0.398
2.25	0.298
1.45	0.238
1	0.198
0.65	0.158
0.25	0.098
0.006	0.048

Table 1: Chemical kinetics of the observed homogeneous reaction.

Let's linearize the data at first. Taking the natural logarithm of both sides of the given rate law expression, we get,

$$\ln(-r) = \ln(k) + n\ln(C)$$

Let, $z = \ln(-r)$, $w = \ln(C)$, and $a_0 = \ln(k)$; implying that $k = e^{a_0}$ and $n = a_1$. The straight line that we end up with is

$$z = a_0 + a_1 w$$

Using the calculator, we obtain the following values for the $\hat{n} = 7$ transformed data points.

$$\sum_{i=1}^{\hat{n}} w_i = -4.3643, \sum_{i=1}^{\hat{n}} z_i = -12.391, \sum_{i=1}^{\hat{n}} w_i z_i = 16.758, \sum_{i=1}^{\hat{n}} w_i^2 = 30.998$$

Now, let's apply the formulae for determining the corresponding linear regression parameters a_0 and a_1.

$$a_1 = \frac{\hat{n}\sum_{i=1}^{\hat{n}} w_i z_i - \sum_{i=1}^{\hat{n}} w_i \sum_{i=1}^{\hat{n}} z_i}{\hat{n}\sum_{i=1}^{\hat{n}} w_i^2 - \left(\sum_{i=1}^{\hat{n}} w_i\right)^2} = 0.31943,\ a_0 = \frac{\sum_{i=1}^{\hat{n}} z_i}{\hat{n}} - a_1\frac{\sum_{i=1}^{\hat{n}} w_i}{\hat{n}} = -1.5711$$

Hence, we obtain

$$\begin{aligned} k &= e^{-1.5711} \\ &= 0.20782 \\ n &= 0.31941 \end{aligned}$$

Plugging these values into the given Power model equation, we get,

$$-r = 0.20782 \times C^{0.31941}$$

Rubric:

- 1 point for the correct linear representation.
- 1 point for the correct a_1 formula.
- 0.5 point for the correct a_1 value.
- 1 point for the correct a_0 formula.
- 0.5 point for the correct a_0 value.
- 0.5 point for the correct k value.

- 0.5 point for the correct n value.

4. Geometrically derive the integral approximation formula of the Single-segment Trapezoidal rule. 2 (CO4) (PO1)

Solution:

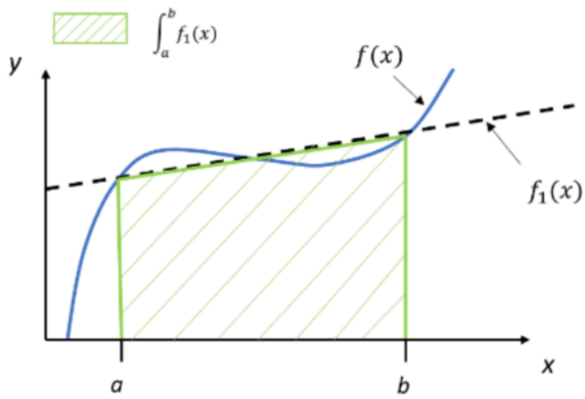

Figure 1: Approximating the function by a first-order polynomial to derive the trapezoidal rule.

We'll approximate the area under the curve $f(x)$ using the area under the straight line $f_1(x)$ within the given interval $[a, b]$, which is essentially a trapezoid (shaded with green color). The integral is –

$$\begin{aligned} I = \int_a^b f(x) \approx \int_a^b f_1(x) &= \text{Area of the trapezoid} \\ &= \frac{1}{2} \times (\text{Sum of parallel sides}) \times (\text{Distance between the parallel sides}) \\ &= \frac{1}{2}[f(b) + f(a)] \times (b - a) \\ &= (b - a)\left[\frac{f(a) + f(b)}{2}\right] \end{aligned}$$

which is the desired integral approximation formula for the single-segment trapezoidal rule.

Rubric:

- 2 points for a correct geometrical derivation.